

Antioxidant intake and primary open-angle glaucoma: a prospective study.



The relation between dietary antioxidant intake and primary open-angle glaucoma risk was examined in participants aged over 40 years in the Nurses' Health Study (n = 76,200) and the Health Professionals Follow-up Study (n = 40,284). They were followed biennially from 1980 and 1986, respectively, to 1996, during periods when they received an eye examination. Dietary intakes were measured repeatedly from 1980 in the Nurses' Health Study and from 1986 in the Health Professionals Follow-up Study using validated food frequency questionnaires. The authors analyzed 474 self-reported glaucoma cases confirmed by medical chart review to have primary open-angle glaucoma with visual field loss. The authors used Cox proportional hazards models for cohort-specific multivariate analyses, and results were pooled using random effects models. The pooled multivariate rate ratios for primary open-angle glaucoma comparing the highest versus lowest quintile of cumulative updated intake were 1.17 (95% confidence interval (CI): 0.87, 1.58) for alpha-carotene, 1.10 (95% CI: 0.82, 1.48) for beta-carotene, 0.95 (95% CI: 0.70, 1.29) for beta-cryptoxanthin, 0.82 (95% CI: 0.60, 1.12) for lycopene, 0.92 (95% CI: 0.69, 1.24) for lutein/zeaxanthin, 1.05 (95% CI: 0.59, 1.89) for vitamin C, 0.97 (95% CI: 0.62, 1.52) for vitamin E, and 1.11 (95% CI: 0.82, 1.51) for vitamin A. In conclusion, the authors did not observe any strong associations between antioxidant consumption and the risk of primary open-angle glaucoma.